

Mohammed Ibrahim

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Summary

Data Scientist with a quantitative engineering foundation and hands-on experience in **applied statistical modeling**, **machine learning**, and **data visualization**. Skilled in **translating complex, multi-source datasets into clear, actionable insights for technical and non-technical audiences**. Experienced building **end-to-end analytics pipelines and dashboards** that support **evidence-based decision-making**. Committed to **fairness, data integrity, and responsible use of predictive models in public-interest and regulatory contexts**.

Key Skills

Languages	Python, R, SQL, HTML, CSS, JavaScript
Statistical Modeling	GLM, Ordinal Logistic Regression, Bayesian Modeling (RStan), A/B Testing, MCMC, Time-series Analysis
ML & NLP	Supervised & Unsupervised Learning, Scikit-Learn, BERTopic, CNNs, Regression, Classification
Visualization & BI	Power BI, Plotly, Altair, Shiny, Matplotlib, Quarto, Dashboard Development
Data & Analytics	ETL Pipelines, Fairness & Bias Evaluation, Peer Code Review, Report Generation for Non-technical Audiences
Databases	PostgreSQL, MongoDB, SQL (large-scale datasets)
Infrastructure	CI/CD, Big Data, Cloud Computing (AWS EC2, S3, EMR Spark)
Tools	GitHub, Docker, VSCode, RStudio, Quarto

Education

Master of Data Science — University of British Columbia

Fully Funded by Mastercard Foundation Scholars Program | Vancouver, Canada Sep 2025 – Jun 2026

- Relevant Coursework: Machine Learning, Data Visualization, Applied Statistics, Statistical Inference, Generalized Linear Models (GLM), Bayesian Methods, Data Science Workflows, Cloud Computing

BSc Electrical and Electronics Engineering (Honors) — University of Khartoum

Concentration: Software Engineering | Khartoum, Sudan Oct 2015 – Dec 2021

- Relevant Coursework: Networks, AI, Cybersecurity, Engineering Economics, Electromagnetic, Database Systems

Projects

Bridging the Gap: LLM Usage Among Data Science Practitioners (Capstone)

University of British Columbia Apr 2026 – Jun 2026

- Designed and analyzed a 136-respondent survey on LLM adoption across data science roles, cleaning and processing a 44-variable dataset using Python
- Applied BERTopic NLP clustering pipeline to free-text responses, ensuring reproducibility with fixed random seeds and documented methodology
- Built a comprehensive suite of visualization scripts (horizontal bar charts, Likert scale plots, heatmaps) to communicate findings to non-technical audiences
- Produced policy-relevant insights on AI tool usage patterns, translating statistical findings into clear written reports and infographics

RAG-Based Product Search System

University of British Columbia Mar 2026 – Apr 2026

- Built a retrieval-augmented generation (RAG) search pipeline over Amazon Reviews data using BM25, FAISS semantic search, and Reciprocal Rank Fusion

- Integrated LLaMA-3 via HuggingFace; deployed **Python Shiny frontend** on Posit Connect; evaluated performance using **precision@k and recall@k metrics**
- Resolved complex dependency conflicts, ensuring **consistent and reproducible output across deployment environments**

Interactive Uber 2024 Performance Dashboard

University of British Columbia

Feb 2026 – Mar 2026

- Developed an **end-to-end interactive dashboard** to visualize annual performance metrics and KPIs for company stakeholders
- **Built a natural-language chat feature** allowing users to request customized visualizations, making data exploration intuitive for **non-technical audiences**
- Applied **Python (Plotly, Shiny)** with ibis/DuckDB backend; designed for **scalable, evidence-based decision-making across stakeholder groups**

AWS Cloud ETL Pipeline & Spark ML (Rainfall Prediction)

University of British Columbia

Jan 2026 – Feb 2026

- Designed and deployed an **ETL pipeline on AWS EC2/S3**, processing **large-scale climate datasets** and orchestrating model training via **EMR Spark (MLlib)**
- Deployed a **REST API endpoint** for model inference; managed cloud credentials, SCP transfers, and environment configuration on Windows

Bank Marketing Prediction

University of British Columbia

Nov 2025 – Dec 2025

- Developed a **classification model** to predict term deposit subscriptions; selected final model after **benchmarking multiple algorithms on accuracy metrics**
- Applied **feature engineering** and polynomial feature selection; implemented **reproducible analysis pipelines** using Quarto and Docker

DataFrame Optimizer Python Package

University of British Columbia

Dec 2025 – Jan 2026

- Built a **dataframe optimization package** reducing memory usage through safe type inference; published to PyPI with CI/CD workflow and automated documentation
- **Conducted peer code reviews and enforced best practices for code quality and reporting consistency across the team**

Professional Experience

Intern Data Scientist — International Institute of Tropical Agriculture (IITA)

Nairobi, Kenya

Aug 2026 – Nov 2026

- Integrating **AgWise agricultural recommendation platform x Limpopo Digital Twin** by designing a **framework** connecting an agronomic decision support tool with a hydrological digital twin's API for **field-level water and rainfall context**
- Training and validating **machine learning models** on agricultural datasets, producing **performance reports and architecture documentation**
- Working on **fine-tuning LLM/Agentic AI** components for an agricultural data corpus, contributing to a **benchmarked prototype and demo interface**
- Supporting development of **data-flow diagrams and use-case definitions** for a functional integrated prototype, including **API/interoperability modules**

Instrumentation Engineer — MT Industrial Supplies and Services

Khartoum, Sudan & South Sudan

Dec 2022 – Oct 2024

- Analyzed **high-frequency flow, pressure, and temperature time-series data** from industrial sensors to **validate system performance and ensure measurement integrity** across large-scale pipeline operations
- Identified **abnormal sensor behavior**, supporting **data-driven troubleshooting** of industrial measurement systems

- **Produced structured daily reports, SAT documentation, and performance handover reports for senior management and regulatory stakeholders**, translating technical findings into clear written communications
- Led instrumentation commissioning for the Block 5A rehabilitation project in Tharjath, South Sudan, contributing to a **30% boost in production output**

Solar Engineer — Freelance

Khartoum, Sudan

Dec 2021 – Oct 2024

- Delivered solar electrification projects generating over 436 kWh of clean energy, serving 200+ people across River Nile and Red Sea states
- **Evaluated solar system performance through load estimation, generation capacity analysis, and reliability modeling** to support **data-informed sizing decisions** for off-grid applications
- Delivered solar power for five farms, replacing unreliable grid electricity and achieving a **25% boost in agricultural production**

Leadership & Volunteer

Volunteer — World Food Programme (WFP)

Port Sudan, Sudan

Apr 2024 – Jul 2024

- **Produced data-driven assessments of essential goods requirements for vulnerable populations**; coordinated distribution logistics using **quantitative demand analysis**

Volunteer Teacher

Northern State, Sudan

Oct 2023 – Jan 2024

- **Delivered structured lessons maintaining learning continuity under challenging conditions** in Al Barkal during displacement